

Code-Layout, Readability and Reusability

Date	01 October 2026
Team ID	SWTID-2026-3533
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the ComicCraft codebase in terms of its structure, readability, and reusability. The project uses a modular FastAPI backend, separate AI-generation modules, Jinja2 templates, and PDF/image processing components so that the application can be maintained and extended.

Code Layout Checklist:

S.No.	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are used throughout the Python, HTML and CSS code.	Yes	Improves readability.
2	Proper File Structure	Files are separated into backend, AI modules, templates, static assets and export-related components.	Yes	Logical modular structure.
3	Meaningful Variable Names	Functions and variables such as generate_outline, generate_story and save_pdf reflect their purpose.	Yes	Names are descriptive.
4	Function / Method Names	Functions are named according to the task they perform, such as generate_image and build_comic_layout.	Yes	Easy to understand.
5	Code Comments	Comments and documentation can explain important AI, routing and export logic.	Partial	Can be expanded where needed.
6	Modular Design	Comic generation, image generation, layout building and PDF export are handled by separate modules.	Yes	Supports reuse and maintenance.
7	No Redundant Code	Common processing is kept in dedicated functions instead of repeating the same workflow.	Yes	Reduces duplication.
8	Error Handling	FastAPI HTTPException and try-except handling are used for unexpected API or processing errors.	Yes	Graceful failure handling.

Reusable Components / Modules:

S.No.	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	FastAPI Routes / routes.py	Python / FastAPI	Form and JSON comic-generation workflows	High
2	Gemini Story Modules	Python / Gemini API	Comic outline, narration and dialogue generation	High
3	Image Generator	Python / Diffusers / Stable Diffusion	Each comic panel requiring an illustration	High
4	Jinja2 Templates	HTML / CSS / Jinja2	Homepage, comic preview and export-success pages	High
5	Layout & PDF Export	Python / FPDF	Combining panel content and exporting the final comic	High

Team Work Allocation:

S.No.	Team Member	Assigned Work
1	C. Iswarya	Project planning, requirements analysis, model selection and overall system architecture.
2	Hasini.D	FastAPI backend, routes.py, user-input processing, API workflow and backend error handling.
3	Ramya.A	Gemini AI integration, comic outline generation, narration/dialogue generation and prompt design.
4	Nandhini	Frontend development using HTML/CSS/Jinja2, input form, comic preview and success pages.
5	Nivetha.K	Stable Diffusion image generation, comic layout, PDF export, testing and final documentation.

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Modular separation of FastAPI routes, AI generation, templates, layout and export components.
Readability	4	Descriptive functions and a clear workflow make the project easy to follow.
Reusability	4	Core generation and export functions can be reused across comic-generation requests.
Documentation / Comments	4	Project documentation describes the modules and workflow; additional inline comments can be added where useful.
Overall Score	4	Good modular organization with scope for further documentation and refinement.

Coding & Solution

Date	5 October 2026
Team ID	SWTID-2026-3533
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Coding & Solution

This section describes the implementation of ComicCraft, an AI-powered comic story creator that uses Gemini models to generate creative comic stories and support the creation of comic content. The solution focuses on a simple user experience, AI-assisted story generation, and organized comic output.

Instructions

1. Ensure the application code is well-structured, readable, and follows suitable development practices.
2. The solution should address the core problem of creating comic stories with AI assistance.
3. Include required source files, configuration files, dependencies, and setup instructions.
4. Maintain the project in a version-control repository such as GitHub.

Solution Summary

Field	Details
Repository Link / URL	GitHub repository for the ComicCraft project
Programming Language(s)	Python / JavaScript (as applicable to the implementation)
Framework(s) Used	Web application framework and Gemini API integration
Key Features Implemented	AI comic story generation; user story/prompt input; Gemini model integration; structured comic-story output; user-friendly interface.
Pending / Incomplete Features	Final deployment, advanced comic-panel customization, and additional enhancements may be completed based on project scope.
Setup / Run Instructions	Install project dependencies, configure the Gemini API key securely, run the application, and open the provided local application URL.

Team Work Allocation

S.No	Name	Assigned Work	Responsibilities
1	C. Iswarya	Project Lead & Requirement Analysis	Coordinate the team, understand project requirements, plan modules, track progress, and support final integration.
2	Hasini.D	AI / Gemini Integration	Work on Gemini model/API integration, prompt handling, AI-generated story logic, and recommendation of suitable prompts.
3	Ramya.A	Story & Comic Generation Module	Develop the story-generation workflow, comic scene/panel structure, character/story details, and output formatting.
4	Nandhini	UI/UX & Front-End Development	Design the user interface, input screens, comic display, navigation, and overall user experience.
5	Nivetha.K	Testing, Documentation & Integration	Test features, identify and report bugs, support module integration, prepare documentation, and assist with final presentation.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / components	Yes
2	Meaningful variable, function, and component names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments

- Team ID: SWTID-2026-3533
- Project: ComicCraft - AI Comic Story Creator using Gemini Models
- Team members: C. Iswarya, Hasini.D, Ramya.A, Nandhini, Nivetha.K
- Work has been allocated according to the order of names provided by the team.

No. of Functional Features Included in the Solution

Date	05 October 2026	Team ID	SWTID-2026-3533
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models	Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features included in the ComicCraft project solution. Each feature is described with its purpose, implementation status, module/component, marks, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks	Contribution
1	User Interface	Create a simple interface for entering comic ideas and viewing generated results.	UI	Done	0.5	Makes the application easy to use.
2	Story Prompt Generation	Accept user prompts and prepare structured comic-story content.	AI / Logic	Done	0.5	Forms the core story-creation workflow.
3	AI Story Generation	Use Gemini models to generate comic story text from user input.	Gemini AI	Done	1.0	Provides the main AI-powered functionality.
4	Character Creation	Generate and organize character names, roles, traits, and dialogue.	AI / Logic	Done	0.5	Builds consistent comic characters.
5	Comic Scene / Panel Creation	Convert the generated story into scene or panel-wise comic content.	Comic Module	Done	0.75	Transforms the story into a comic format.
6	Image / Visual Integration	Support visual content for comic scenes where applicable.	Media / AI	In Progress	0.5	Improves the visual storytelling experience.
7	Preview and Edit	Allow users to preview generated content and make basic changes.	UI / Logic	In Progress	0.5	Helps users refine the generated comic.
8	Save / Export	Provide a way to save or export the completed comic output.	Storage / Output	In Progress	0.75	Makes the final comic reusable and shareable.

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	5
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	5

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Comic prompt screen; Preview and Edit
2	Backend / Logic	2	Story prompt processing; Character creation
3	Database / Storage	1	Save / Export
4	API / Integration	2	Gemini model integration; AI content generation
5	Security / Authentication	1	Basic input validation / controlled access (if enabled)

Team Members & Work Allocation

S.No	Team Member	Assigned Work
1	C. Iswarya	Project planning, requirement analysis, and overall coordination
2	Hasini.D	User interface design and frontend implementation
3	Ramya.A	Gemini AI integration and story-generation logic
4	Nandhini	Character creation, comic scene/panel structure, and content organization
5	Nivetha.K	Testing, debugging, documentation, and final output/export

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Note: Work is assigned in the same order as the names provided.